

Task21

Name: Aditya Santosh Nivangune

College: Zeal Polytechnic

Domain/Branch: AIML

Q1.

```
# Streamlit is used to create the web application
import streamlit as st

# Pandas is used for creating and manipulating DataFrames
import pandas as pd

# Joblib is used for loading saved ML model and preprocessing objects
import joblib
```

Q2.

```
# Load trained Linear Regression model
model = joblib.load("LR_ford_car.pkl")

# Load StandardScaler
scaler = joblib.load("scaler.pkl")

# Load encoded column names
encoded_columns = joblib.load("columns.pkl")

# Load saved files
model = joblib.load("LR_ford_car.pkl")
scaler = joblib.load("scaler.pkl")
encoded_columns = joblib.load("columns.pkl")
```

Q3.

```
# Configure the Streamlit page settings.
# page_title sets the title shown on the browser tab.
# layout="centered" keeps the content centered for a clean and user-friendly interface.
# Page configuration
st.set_page_config(
    page_title="Ford Car Price Predictor",
    layout="centered"
)
```

Q4.

```

# Title
st.title("Ford Car Price Predictor")

# Description card
st.write(
    """
    <div class="description">
    Predicting resale prices using advanced data analytics
    </div>
    """,
    unsafe_allow_html=True
)

# Footer
st.write(
    """
    <br>
    <center>
    Machine Learning Based Vehicle Valuation System
    </center>
    """,
    unsafe_allow_html=True)

```

Q5.

```

# Input section
st.subheader("Enter Vehicle Details")
col1, col2 = st.columns(2)

with col1:
    model_name = st.text_input(
        "Car Model")

    year = st.number_input(
        "Manufacturing Year",
        min_value=1990,
        max_value=2025,
        value=2018 )

    mileage = st.number_input(
        "Mileage",
        min_value=0,
        max_value=300000,
        value=50000 )

    tax = st.number_input(
        "Road Tax",
        min_value=0,
        max_value=600,
        value=150 )

```

Q6.

```

with col2:
    transmission = st.selectbox(
        "Transmission",
        [
            "Automatic",
            "Manual",
            "Semi-Auto" ])

    fuelType = st.selectbox(
        "Fuel Type",
        [
            "Petrol",
            "Diesel",
            "Hybrid",
            "Electric",
            "Other" ])

    mpg = st.number_input(
        "MPG",
        min_value=0.0,
        max_value=100.0,
        value=50.0 )

    engineSize = st.number_input(
        "Engine Size",
        min_value=0.5,
        max_value=6.0,
        value=1.5 )

```

Q7.

```

model_name = st.text_input(
    "Car Model")

# Prediction button
predict = st.button("Predict Price")

```

Q8.

```

if predict:
    try:
        # Create input dataframe
        data = pd.DataFrame({
            "model": [model_name],
            "year": [year],
            "transmission": [transmission],
            "mileage": [mileage],
            "fuelType": [fuelType],
            "tax": [tax],
            "mpg": [mpg],
            "engineSize": [engineSize] })

        # One hot encoding
        data = pd.get_dummies(data)

        # Match training columns
        data = data.reindex(
            columns=encoded_columns,
            fill_value=0
        )

```

Q9.

```

# Scale numerical values
numerical_columns = [
    "year",
    "mileage",
    "tax",
    "mpg",
    "engineSize"
]
data[numerical_columns] = scaler.transform(
    data[numerical_columns]
)

# Prediction
prediction = model.predict(data)

# Display result
st.success(
    f"Estimated Selling Price: £{prediction[0]:.2f}"
)
except Exception as e:
    st.error(
        f"Prediction Error: {e}"
    )

```

Q.10

Ford Car Price Predictor

Predicting resale prices using advanced data analytics

Enter Vehicle Details

Car Model	Transmission
Fiesta	Automatic
Manufacturing Year	Fuel Type
2020	Petrol
Mileage	MPG
50000	56.00
Road Tax	Engine Size
130	1.50

Predict Price

Estimated Selling Price: £11,875.17

Machine Learning Based Vehicle Valuation System